

The sharp-interface and phase-field cell problems are the same problem

1 Setup

A periodic cell Ω splits into two subdomains meeting on an interface Γ ,

$$\Omega = \Omega_1 \cup \Omega_2, \quad \Omega_1 \cap \Omega_2 = \emptyset, \quad \Gamma = \partial\Omega_1 \cap \partial\Omega_2,$$

with constant conductivities $K_1, K_2 > 0$ and $\hat{\mathbf{n}}$ the unit normal on Γ pointing out of Ω_1 . Homogenization gives the effective tensor $\mathbf{K}_{\text{eff}}$ from a *corrector* (cell) problem posed on Ω . There are two ways to write that problem, and this note shows they are the same one. Nothing below uses any property of the phases beyond K_1, K_2 being positive constants.

Notation. χ is the *indicator function* of Ω_1 : the function equal to 1 inside Ω_1 and 0 outside,

$$\chi(\mathbf{x}) = \mathbf{1}_{\Omega_1}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \Omega_1, \\ 0, & \mathbf{x} \in \Omega_2. \end{cases} \quad (1)$$

φ is the phase field: a smooth function equal to 1 in Ω_1 and 0 in Ω_2 , sweeping between them across a band of decay length ε . It is a smoothed χ . $\langle \cdot \rangle = |\Omega|^{-1} \int_{\Omega}(\cdot)$, and $\llbracket \cdot \rrbracket$ is the jump across Γ .

Problem A (sharp). Find *two* cell-periodic fields, $\mathbf{t}_1$ defined on Ω_1 only and $\mathbf{t}_2$ defined on Ω_2 only, with

$$\nabla \cdot (K_1(\nabla \mathbf{t}_1 + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_1, \quad \nabla \cdot (K_2(\nabla \mathbf{t}_2 + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_2, \quad (2)$$

$$\mathbf{t}_1 = \mathbf{t}_2 \text{ on } \Gamma, \quad K_1(\nabla \mathbf{t}_1 + \mathbf{I}) \cdot \hat{\mathbf{n}} = K_2(\nabla \mathbf{t}_2 + \mathbf{I}) \cdot \hat{\mathbf{n}} \text{ on } \Gamma, \quad (3)$$

normalized by $\int_{\Omega_1} \mathbf{t}_1 d\Omega + \int_{\Omega_2} \mathbf{t}_2 d\Omega = \mathbf{0}$, and then

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \left[\int_{\Omega_1} K_1(\nabla \mathbf{t}_1 + \mathbf{I}) d\Omega + \int_{\Omega_2} K_2(\nabla \mathbf{t}_2 + \mathbf{I}) d\Omega \right]. \quad (4)$$

The two interface conditions are the physics: the field is continuous, and the normal flux is conserved. This is Calonne et al. (2011), Eqs. (7)–(11), whose $\mathbf{t}_i$ (ice) and $\mathbf{t}_a$ (air) are our $\mathbf{t}_1$ and $\mathbf{t}_2$. In the jump notation, (3) reads $\llbracket \mathbf{t} \rrbracket = \mathbf{0}$ and $\llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = \mathbf{0}$, where the jump of a two-field quantity means (side 1)–(side 2). Note that A has two unknowns, B below has one; reconciling that is part of the claim (§2(f)).

Problem B (phase field). Find cell-periodic $\mathbf{t}$ with

$$\nabla \cdot (K(\varphi)(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega, \quad \langle \mathbf{t} \rangle = \mathbf{0}, \quad K(\varphi) = K_2 + (K_1 - K_2)\varphi, \quad (5)$$

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \int_{\Omega} K(\varphi)(\nabla \mathbf{t} + \mathbf{I}) d\Omega. \quad (6)$$

One domain, one equation, no interface conditions.

Claim. A and B are the same problem. Two steps: §2 collapses A's two domains and two correctors into one weak equation for one field; §3 shows A's coefficient *is* B's coefficient. Neither step is a limit.

2 Step 1: two domains and two correctors become one weak equation

Let $v \in H_{\text{per}}^1(\Omega)$ be an arbitrary cell-periodic scalar test function. We treat each subdomain separately, then add.

(a) Test each bulk equation on its own subdomain. Multiply the first equation of (2) by v and integrate over Ω_1 ; do the same on Ω_2 :

$$\int_{\Omega_1} v \nabla \cdot (K_1(\nabla \mathbf{t}_1 + \mathbf{I})) d\Omega = 0, \quad \int_{\Omega_2} v \nabla \cdot (K_2(\nabla \mathbf{t}_2 + \mathbf{I})) d\Omega = 0. \quad (7)$$

Both hold because the integrands vanish pointwise. No approximation yet.

(b) Integrate by parts on each subdomain. Using $\int_D v \nabla \cdot \mathbf{A} = -\int_D \nabla v \cdot \mathbf{A} + \oint_{\partial D} v \mathbf{A} \cdot \hat{\mathbf{n}}_D$ with $\hat{\mathbf{n}}_D$ the outward normal of D . The boundary of Ω_1 consists of Γ , where its outward normal is $+\hat{\mathbf{n}}$, and whatever part of the outer cell boundary $\partial\Omega$ it touches; likewise for Ω_2 , whose outward normal on Γ is $-\hat{\mathbf{n}}$. Writing $\mathbf{A}_m = K_m(\nabla \mathbf{t}_m + \mathbf{I})$ for $m = 1, 2$,

$$0 = -\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 d\Omega + \int_{\Gamma} v \mathbf{A}_1 \cdot \hat{\mathbf{n}} d\Gamma + \underbrace{\int_{\partial\Omega \cap \partial\Omega_1} v \mathbf{A}_1 \cdot \hat{\mathbf{n}}_{\partial\Omega} dS}_{O_1}, \quad (8)$$

$$0 = -\int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 d\Omega - \int_{\Gamma} v \mathbf{A}_2 \cdot \hat{\mathbf{n}} d\Gamma + \underbrace{\int_{\partial\Omega \cap \partial\Omega_2} v \mathbf{A}_2 \cdot \hat{\mathbf{n}}_{\partial\Omega} dS}_{O_2}. \quad (9)$$

The sign flip on the Γ term of (9) is the only subtlety: Γ is traversed with opposite orientation from the two sides.

(c) Add the two equations. This is the step that produces a single statement on Ω . Adding (8) and (9),

$$0 = -\left[\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 d\Omega + \int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 d\Omega \right] + \int_{\Gamma} v (\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} d\Gamma + O_1 + O_2. \quad (10)$$

Now dispose of the three groups in turn.

(d) The outer boundary terms cancel. $O_1 + O_2$ is an integral over all of $\partial\Omega$. Both v and the fluxes $\mathbf{A}_1, \mathbf{A}_2$ are cell-periodic, so they take equal values at corresponding points of opposite faces, where $\hat{\mathbf{n}}_{\partial\Omega}$ points in opposite directions. Every contribution is therefore matched by its negative: $O_1 + O_2 = 0$.

(e) The interface term is exactly the flux jump, which vanishes. The middle term of (10) is $\int_{\Gamma} v (\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} d\Gamma$, and $(\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} = 0$ on Γ is exactly the second interface condition in (3). So the term is 0 for every v , since the integrand vanishes pointwise.

(f) **The two volume integrals merge into one.** This is where $K_\star$ and the single corrector $\mathbf{t}$ come from. For both it is the same bookkeeping, with one extra check for $\mathbf{t}$. Define the piecewise-constant coefficient and the glued corrector

$$K_\star := K_1\chi + K_2(1 - \chi), \quad \mathbf{t} := \mathbf{t}_1\chi + \mathbf{t}_2(1 - \chi), \quad \text{so} \quad (K_\star, \mathbf{t}) = \begin{cases} (K_1, \mathbf{t}_1) & \text{on } \Omega_1, \\ (K_2, \mathbf{t}_2) & \text{on } \Omega_2. \end{cases} \quad (11)$$

Each piece of $\mathbf{t}$ is only ever evaluated on its own subdomain, so $\mathbf{t}_1$ never needs a value in Ω_2 or vice versa.

The coefficient. $K_\star$ equals K_1 at every point of Ω_1 and K_2 at every point of Ω_2 . Nothing more is needed: a coefficient may jump, and $K_\star$ does.

The corrector. The extra check concerns the gradient. On each open subdomain $\nabla \mathbf{t} = \nabla \mathbf{t}_m$, so the *piecewise* gradient of $\mathbf{t}$ is $\chi \nabla \mathbf{t}_1 + (1 - \chi) \nabla \mathbf{t}_2$. The question is whether this is also the gradient of $\mathbf{t}$ *as a function on* Ω , i.e. its weak gradient, which is what a weak form on Ω differentiates. Test against a smooth periodic vector field $\boldsymbol{\psi}$ and integrate by parts on each subdomain exactly as in (b), with the same orientations and the same periodic cancellation as (d):

$$\int_{\Omega} \mathbf{t} \cdot \nabla \boldsymbol{\psi} \, d\Omega = - \int_{\Omega_1} \nabla \mathbf{t}_1 \cdot \boldsymbol{\psi} \, d\Omega - \int_{\Omega_2} \nabla \mathbf{t}_2 \cdot \boldsymbol{\psi} \, d\Omega + \int_{\Gamma} (\mathbf{t}_1 - \mathbf{t}_2) \boldsymbol{\psi} \cdot \hat{\mathbf{n}} \, d\Gamma. \quad (12)$$

The last term vanishes for every $\boldsymbol{\psi}$ *if and only if* $\mathbf{t}_1 = \mathbf{t}_2$ on Γ , the first interface condition in (3). When it does, $\mathbf{t} \in H_{\text{per}}^1(\Omega)$ and its weak gradient is the piecewise one:

$$\nabla \mathbf{t} = \nabla \mathbf{t}_1 \text{ on } \Omega_1, \quad \nabla \mathbf{t} = \nabla \mathbf{t}_2 \text{ on } \Omega_2, \quad \text{with no part concentrated on } \Gamma. \quad (13)$$

Had the fields jumped, $\nabla \mathbf{t}$ would carry an extra surface term $[[\mathbf{t}]] \hat{\mathbf{n}} \delta_{\Gamma}$. The piecewise integrals of (10) would not see it, but the weak form on Ω would, so the two would no longer agree. This is the only place where $K_\star$ and $\mathbf{t}$ behave differently. The coefficient may be discontinuous; the corrector must not be, and continuity is exactly what permits the gluing.

The merge. With (11) and (13) we may substitute on each subdomain without changing either integral:

$$\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 + \int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 = \int_{\Omega_1} \nabla v \cdot K_\star(\nabla \mathbf{t} + \mathbf{I}) + \int_{\Omega_2} \nabla v \cdot K_\star(\nabla \mathbf{t} + \mathbf{I}). \quad (14)$$

The two integrands are now the *same function* of $\mathbf{x}$, integrated over two disjoint sets whose union is Ω . Additivity of the integral gives

$$\int_{\Omega_1} (\cdots) + \int_{\Omega_2} (\cdots) = \int_{\Omega_1 \cup \Omega_2} (\cdots) = \int_{\Omega} (\cdots), \quad (15)$$

the last equality because $\Omega_1 \cup \Omega_2$ differs from Ω only by Γ , a set of measure zero. So neither $K_\star$ nor $\mathbf{t}$ is assumed or fitted. Each is the name for “one thing on one piece and another on the other,” and (15) is the only thing done with them. The same additivity turns $\mathbf{A}$ ’s normalization into $\langle \mathbf{t} \rangle = \mathbf{0}$ and $\mathbf{A}$ ’s $\mathbf{K}_{\text{eff}}$ (4) into $|\Omega|^{-1} \int_{\Omega} K_\star(\nabla \mathbf{t} + \mathbf{I})$.

(g) **Result.** Substituting (d), (e), (f) into (10):

$$\boxed{\int_{\Omega} \nabla v \cdot K_\star(\nabla \mathbf{t} + \mathbf{I}) \, d\Omega = 0 \quad \forall v \in H_{\text{per}}^1(\Omega).} \quad (16)$$

Remark 1 (Where the two interface conditions went). *They were not dropped; each was absorbed somewhere. Flux conservation $\llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = 0$ was used in step (e), and it is the natural condition of (16): run (e) backwards and any weak solution satisfies it automatically. Continuity $\mathbf{t}_1 = \mathbf{t}_2$ was used in step (f), to glue the two correctors into one $\mathbf{t} \in H_{\text{per}}^1(\Omega)$ via (12). Conversely, it is built into (16): a single H^1 field on Ω cannot jump, so restricting any weak solution to Ω_1 and Ω_2 returns a pair $(\mathbf{t}_1, \mathbf{t}_2)$ with matching traces. Two correctors in A and one in B is therefore not a difference between the problems. A's pair is B's single field, written in two pieces.*

Remark 2 (Why one arbitrary v on Ω is not an extra assumption). *It may look as though testing with a single v defined on all of Ω smuggles in something. It does not, and the step reverses: given (16), choose v supported strictly inside Ω_1 . Then the integral is over Ω_1 only, where $K_\star = K_1$ and $\mathbf{t} = \mathbf{t}_1$; undoing the integration by parts (no boundary term, since v vanishes near $\partial\Omega_1$) returns the first equation of (2). Symmetrically for Ω_2 . Allowing v to straddle Γ then returns the interface term, hence (3). So (16) is equivalent to (2)–(3), not weaker.*

Remark 3 (Same algebra for the physical field). *Nothing in (a)–(g) used the specific form $\nabla \mathbf{t} + \mathbf{I}$. Replacing it by ∇T throughout collapses the physical problem, with its own two temperatures T_1 on Ω_1 and T_2 on Ω_2 , $\nabla \cdot (K_m \nabla T_m) = 0$, $T_1 = T_2$ and $K_1 \nabla T_1 \cdot \hat{\mathbf{n}} = K_2 \nabla T_2 \cdot \hat{\mathbf{n}}$ on Γ , into $\int_\Omega \nabla v \cdot K_\star \nabla T = 0$ for the glued T , identically.*

3 Step 2: the coefficient is already the phase-field interpolation

It is tempting to finish by saying “now replace $K_\star$ by a smooth $K(\varphi)$ that converges to it.” That would be an approximation, and would need a convergence theorem. No replacement is needed.

Lemma 1. *Let $K(\cdot)$ be any function with $K(0) = K_2$ and $K(1) = K_1$. Then*

$$K(\chi) = K_\star \quad \text{pointwise on } \Omega, \quad (17)$$

exactly — no limit, no smallness of ε .

Proof. χ takes only the values 0 and 1. At $\mathbf{x} \in \Omega_1$, $\chi = 1$, so $K(\chi) = K(1) = K_1 = K_\star$. At $\mathbf{x} \in \Omega_2$, $\chi = 0$, so $K(\chi) = K(0) = K_2 = K_\star$. Those are all the points of Ω . $\square$

For the affine K of (5) it is three lines:

$$K_\star \stackrel{(11)}{=} K_1 \chi + K_2 (1 - \chi) = K_2 + (K_1 - K_2) \chi = K(\chi). \quad (18)$$

What this means. Problems A and B do not have two constitutive laws that agree asymptotically. They have *one* law $K(\cdot)$, evaluated at two different arguments:

$$\text{A: } K(\chi), \quad \text{B: } K(\varphi).$$

So Step 2 approximates the *geometry*, never the equations, and every difference between A and B traces to the single object $\varphi - \chi$. A second consequence: since χ never takes an intermediate value, the *shape* of K between 0 and 1 is invisible to Problem A. Every admissible interpolation — arithmetic, harmonic, any other — coincides exactly on χ . The choice can therefore matter only through the diffuse band (§5).

Remark 4 (Why the corrector needs no Step 2). $K_\star$ and $\mathbf{t}$ are glued the same way in (f), but only $K_\star$ reappears here. There is no “ $\mathbf{t}(\varphi)$ ” interpolating $\mathbf{t}_1$ and $\mathbf{t}_2$ across the band, and none is needed. K is data: B has to be told what coefficient to use at every point, including inside the band, and the lemma shows its prescription $K(\varphi)$ reduces to A ’s on χ . $\mathbf{t}$ is the unknown: B solves for one field on all of Ω , and inside the band it simply takes whatever values the equation produces. At finite ε the two sharp pieces $\mathbf{t}_1, \mathbf{t}_2$ have no separate meaning inside B_ε , and nothing requires them to. After Step 1 A also has a single unknown in the same space $H_{\text{per}}^1(\Omega)$, so the two problems differ only in their coefficient, and that difference is carried entirely by $\varphi - \chi$.

Proposition 1. *If $\varphi = \chi$, Problems A and B are the same boundary-value problem and give the same $\mathbf{K}_{\text{eff}}$, identically.*

Proof. Step 1 gives $A \Leftrightarrow (16)$. Lemma 1 gives $K_\star = K(\chi)$, so (16) is the weak form of (5). The two $\mathbf{K}_{\text{eff}}$ formulas agree by (14)–(15) applied to the integrand $K(\nabla \mathbf{t} + \mathbf{I})$. $\square$

4 Finite ε : what is left over

In practice $\varphi \neq \chi$, so Proposition 1 does not apply verbatim. By Lemma 1 the entire residual difference is carried by $\varphi - \chi$. For the affine K this is exact:

$$\delta K := K(\varphi) - K_\star = (K_1 - K_2)(\varphi - \chi), \quad \text{supp } \delta K = B_\varepsilon := \{0 < \varphi < 1\}. \quad (19)$$

Both problems are uniformly elliptic on the same bounds, $\min(K_1, K_2) \leq K \leq \max(K_1, K_2)$, so both are well posed and both admit the Dirichlet minimum principle: for a unit direction $\mathbf{e}$,

$$A(K) := \mathbf{e} \cdot \mathbf{K}_{\text{eff}}[K] \mathbf{e} = \min_{v \in H_{\text{per}}^1(\Omega)} E_K[v], \quad E_K[v] := \langle (\nabla v + \mathbf{e}) \cdot K(\nabla v + \mathbf{e}) \rangle. \quad (20)$$

Write $K_1 \rightarrow K(\varphi)$, $K_2 \rightarrow K_\star$ with minimizers $v_\varepsilon, v_\star$. Here $v_\star$ is the *glued* sharp corrector of (11); it is because gluing puts it in the same space $H_{\text{per}}^1(\Omega)$ as v_ε that each can be tested in the other’s energy. For *any* v , $E_{K(\varphi)}[v] - E_{K_\star}[v] = \langle \delta K |\nabla v + \mathbf{e}|^2 \rangle$. Testing each minimizer against the other’s energy gives two inequalities which combine into

$$\langle \delta K |\nabla v_\varepsilon + \mathbf{e}|^2 \rangle \leq A_\varepsilon - A_\star \leq \langle \delta K |\nabla v_\star + \mathbf{e}|^2 \rangle. \quad (21)$$

Both sides are integrals over B_ε only. The left one uses only the corrector already computed, so (21) is an *a posteriori* error bar, not an asymptotic claim. With the logistic profile $\varphi = \sigma(s/\varepsilon)$, $\sigma(u) = (1 + e^{-u})^{-1}$, in the signed distance s to Γ , and $\int_0^\infty (1 - \sigma) du = \ln 2$,

$$\|\varphi - \chi\|_{L^1(\Omega)} = 2 \ln 2 \, \varepsilon |\Gamma| + \mathcal{O}(\varepsilon^2) \implies |A_\varepsilon - A_\star| \leq 2 \ln 2 \, (K_1 - K_2) M^2 \frac{|\Gamma|}{|\Omega|} \varepsilon + \mathcal{O}(\varepsilon^2), \quad (22)$$

with $M^2 = \max_{B_\varepsilon} |\nabla v + \mathbf{e}|^2$. First order, computable prefactor, and no H-convergence theorem required. (That theorem is the right citation only for a non-affine K , where (19) becomes an inequality.)

5 The first-order error is a surface excess

To first order a thin graded layer is equivalent to a sharp interface carrying two *surface excesses* per unit area. Transport tangential to Γ crosses the layer’s sublayers in parallel and sees an excess conductance; transport normal to it crosses them in series and sees an excess resistance:

$$\Sigma_t = \int_{-\infty}^{\infty} [K(\varphi) - K_\star] ds, \quad \Sigma_n = \int_{-\infty}^{\infty} \left[\frac{1}{K(\varphi)} - \frac{1}{K_\star} \right] ds. \quad (23)$$

These two numbers are the only $\mathcal{O}(\varepsilon)$ data the band presents. Both are elementary for the logistic profile, whose defining property here is the antisymmetry $\sigma(-u) = 1 - \sigma(u)$.

$\Sigma_t = 0$, **exactly**. K is affine, so the integrand is $(K_1 - K_2)[\sigma(u) - H(u)]$ with H the unit step, and antisymmetry makes $\sigma - H$ an odd function:

$$\Sigma_t = (K_1 - K_2) \varepsilon \int_{-\infty}^{\infty} [\sigma(u) - H(u)] du = 0. \quad (24)$$

Not first order — zero. The diffuse cell has *exactly* the sharp cell's phase fraction, so the parallel response carries no error at any ε . The leading correction is $\mathcal{O}(\varepsilon^2 \kappa)$ from interface curvature.

$\Sigma_n \neq 0$. Here $1/K$ is convex in φ , so the band short-circuits a series resistance. With $p = \sigma(u)$, $du = dp/[p(1-p)]$ and partial fractions,

$$\Sigma_n = \varepsilon \int_0^1 \left[\frac{1}{K_2 + (K_1 - K_2)p} - \frac{1}{K_*} \right] \frac{dp}{p(1-p)} = -\varepsilon \left(\frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2} < 0. \quad (25)$$

Negative excess resistance: the diffuse cell conducts too well normal to Γ , first order in ε , with no free constants. This is the quantitative form of the familiar statement that an arithmetic mixture rule is right in parallel and wrong in series.

Corollary 1 (An interpolation with no first-order bias). *The harmonic law $K_h(\varphi) = (\varphi/K_1 + (1-\varphi)/K_2)^{-1}$ has $1/K_h$ affine in φ , so the same antisymmetry gives $\Sigma_n = 0$ — but then $\Sigma_t \neq 0$. No scalar law nulls both. The tensorial law*

$$\mathbf{K}(\varphi) = K(\varphi)(\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_h(\varphi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \nabla \varphi / |\nabla \varphi|, \quad (26)$$

uses each where its excess vanishes, giving $\Sigma_t = \Sigma_n = 0$ and $\mathcal{O}(\varepsilon^2)$. Both branches satisfy $K(0) = K_2$, $K(1) = K_1$, so Lemma 1 and Proposition 1 still apply.

This is available because the conductivity cell problem is post-processing on a frozen φ : its interpolation is independent of any other model, and choosing it does not bear on one. §7 applies it to the cell problem.

6 Worked example and test: the laminate

For layers of period L with phase fraction $\bar{\varphi}$ and n_Γ interfaces per period, $\mathbf{K}_{\text{eff}}$ is diagonal with $k_{\parallel} = \langle K \rangle$ and $k_{\perp} = \langle K^{-1} \rangle^{-1}$. Equations (24)–(25) give

$$k_{\parallel}(\varepsilon) = K_2 + (K_1 - K_2)\bar{\varphi} \quad \text{exact for every } \varepsilon, \quad (27)$$

$$\langle K^{-1} \rangle_{\varepsilon} = \frac{\bar{\varphi}}{K_1} + \frac{1-\bar{\varphi}}{K_2} - \frac{n_\Gamma \varepsilon}{L} \left(\frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2}, \quad k_{\perp} = \langle K^{-1} \rangle_{\varepsilon}^{-1}. \quad (28)$$

Equation (28) is not a truncation: its only error is tail overlap between neighbouring interfaces, $\mathcal{O}(e^{-L/4\varepsilon})$, which against direct quadrature is 5×10^{-5} relative at $\varepsilon = L/50$ and machine precision by $\varepsilon = L/128$.

In our application $K_1/K_2 = 2.29/0.02 = 114.5$, giving $(K_2^{-1} - K_1^{-1}) \ln(K_1/K_2) = 234.959$ and, at $\bar{\varphi} = \frac{1}{2}$, $n_\Gamma = 2$:

ε	$\langle K^{-1} \rangle$	$k_{\perp}$	bias
$L/50$	15.8200	0.063211	+59.4%
$L/128$	21.5471	0.046410	+17.0%
$L/512$	24.3005	0.041151	+3.8%
sharp	25.2183	0.039654	—

How to test it. Equation (28) is *linear in ε in resistivity* and a hyperbola in conductivity. Fit and plot $1/k_{\perp}$ against ε ; then the test has two predicted numbers and no fitted ones,

$$\text{intercept} = \frac{\bar{\varphi}}{K_1} + \frac{1 - \bar{\varphi}}{K_2}, \quad \text{slope} = \frac{n_{\Gamma}}{L} \left(\frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2}. \quad (29)$$

Two ways to get a meaningless answer: evaluating (29) at the nominal rather than the *measured* $\bar{\varphi}$ (the discrete field's mean is not exactly $\bar{\varphi}$), and refining ε at fixed mesh, which confounds this bias with mesh convergence. The committed driver is `studies/keff_sharp_limit/verification/`.

7 Applying the tensor law to the steady-state cell problem

Scope. Everything in this note concerns the *steady-state* corrector problem that defines $\mathbf{K}_{\text{eff}}$ for a frozen microstructure. It says nothing about the transient energy balance of the metamorphism models, which have their own interpolations. Nothing here justifies, or asks for, a change to them.

Strong form. Replace the scalar coefficient of (5) by the tensor (26), written $\mathbf{K}$ below with $K(\varphi)$ the arithmetic law. For each direction m , find cell-periodic t_m with

$$\nabla \cdot (\mathbf{K}(\varphi, \nabla \varphi) (\nabla t_m + \mathbf{e}_m)) = 0 \text{ in } \Omega, \quad \langle t_m \rangle = 0, \quad \mathbf{K}_{\text{eff}} \mathbf{e}_m = \langle \mathbf{K} (\nabla t_m + \mathbf{e}_m) \rangle. \quad (30)$$

With $\mathbf{G} = \nabla t_m + \mathbf{e}_m$, the local flux splits into a part along the interface and a part across it,

$$\mathbf{K} \mathbf{G} = K(\varphi) \mathbf{G} + (K_h(\varphi) - K(\varphi)) (\hat{\mathbf{n}} \cdot \mathbf{G}) \hat{\mathbf{n}}, \quad (31)$$

so only the normal component of the gradient sees the harmonic mean. In the bulk, where $\varphi \in \{0, 1\}$, the correction term vanishes and (30) is the scalar problem of Proposition 1.

Weak form. Multiply by a periodic test function v and integrate by parts. Periodicity removes the boundary term exactly as in step (d):

$$\int_{\Omega} \nabla v \cdot \mathbf{K} \nabla t_m d\Omega = - \int_{\Omega} \nabla v \cdot \mathbf{K} \mathbf{e}_m d\Omega \quad \forall v \in H_{\text{per}}^1(\Omega). \quad (32)$$

The bilinear form on the left does not depend on m ; only the load does.

Well posed, with the same bounds. $\mathbf{K}$ is symmetric, with eigenvalue $K(\varphi)$ on the tangent plane and $K_h(\varphi)$ along $\hat{\mathbf{n}}$. Both lie in $[\min(K_1, K_2), \max(K_1, K_2)]$, so

$$\min(K_1, K_2) |\boldsymbol{\xi}|^2 \leq \boldsymbol{\xi} \cdot \mathbf{K} \boldsymbol{\xi} \leq \max(K_1, K_2) |\boldsymbol{\xi}|^2. \quad (33)$$

By Lax–Milgram, (32) has a unique solution with $\langle t_m \rangle = 0$. The minimum principle (20) holds with K replaced by $\mathbf{K}$, and so does the sandwich (21). Testing (32) for direction i with $v = t_j$ gives the symmetric energy form

$$\mathbf{e}_i \cdot \mathbf{K}_{\text{eff}} \mathbf{e}_j = \langle (\nabla t_i + \mathbf{e}_i) \cdot \mathbf{K} (\nabla t_j + \mathbf{e}_j) \rangle, \quad (34)$$

so $\mathbf{K}_{\text{eff}}$ is symmetric positive definite, as it must be.

Discrete form. With basis functions N_a and $t_m = \sum_b T_{m,b} N_b$, (32) becomes

$$\mathbf{A} \mathbf{T}_m = \mathbf{F}_m, \quad \mathbf{A}_{ab} = \int_{\Omega} \nabla N_a \cdot \mathbf{K} \nabla N_b d\Omega, \quad (\mathbf{F}_m)_a = - \int_{\Omega} \nabla N_a \cdot \mathbf{K} \mathbf{e}_m d\Omega. \quad (35)$$

$\mathbf{A}$ is symmetric positive semidefinite, with the constants as its only null space (removed by the mean constraint). It is assembled once and reused for all directions, and conjugate gradients with algebraic multigrid apply as in the scalar case. $\mathbf{K}$ is evaluated at each quadrature point from φ and $\nabla\varphi$ there.

Worked example: the laminate. Here $\hat{\mathbf{n}} = \pm \mathbf{e}_y$ everywhere, so $\mathbf{K} = \text{diag}(K(\varphi), K_h(\varphi))$ and (30) decouples. Along the layers the corrector is zero and $k_{\parallel} = \langle K(\varphi) \rangle$. Across them the flux is constant, so $k_{\perp} = \langle 1/K_h(\varphi) \rangle^{-1}$. Both averages are of functions affine in φ , so both equal the sharp values at every ε (up to the $\mathcal{O}(e^{-L/4\varepsilon})$ tail overlap of §6):

$$k_{\parallel} = K_2 + (K_1 - K_2)\bar{\varphi}, \quad k_{\perp} = \left(\frac{\bar{\varphi}}{K_1} + \frac{1 - \bar{\varphi}}{K_2} \right)^{-1}. \quad (36)$$

Compare (28): the ε term is gone. Measured, the tensor law reproduces $k_{\perp}$ to 4×10^{-7} from $\varepsilon = L/50$ to $L/512$, where the scalar law is 59% to 3.8% high. The derivation of (26) and a curved-interface test (a disk array against Rayleigh’s solution) are in the companion writeup `tensor_conductivity_law.tex`.

8 Where this lives in the code

The cell problem is solved in-line by `enceladus_dsm (-keff 1)`, or on saved snapshots with `-keff_replay`. The law is chosen with `-keff_interp arith|tensor|sharp`; the default is `tensor` (before 2026-09-23 it was `arith`). Non-default laws write `k_eff_<law>.csv`.

Inputs at each quadrature point (`src/keff_field.c`, `KeffProjectIce`). The phase field and its gradient are evaluated on the solver mesh and stored per quadrature point: `kc->ice[idx] =  $\varphi$` and `kc->grad_ice[idx*dim+d] =  $\partial_d \varphi$` . The gradient array is allocated only for the tensor law.

The coefficient (`src/keff_cell.c`, `KeffPointCond`), row-major $\mathbf{K} = K[a*\text{dim}+b]$:

```
ThermalCond(kc->app, phi, &cond, NULL); /* K(phi), arithmetic */
for (a = 0; a < dim; a++) K[a*dim + a] = cond; /* K(phi) I */
if (kc->interp == KEFF_INTERP_TENSOR) {
    k_harm = 1.0 / (phi/k_ice + (1.0 - phi)/k_air); /* K_h(phi) */
    g2 = |grad phi|^2;
    if (g2 > 0.0) /* + (K_h - K) n(x) n */
        for (a...) for (b...)
            K[a*dim + b] += (k_harm - cond) * g[a]*g[b] / g2;
}
```

This is (26) written as (31): $K \mathbf{I} + (K_h - K) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}$ with $\hat{\mathbf{n}} \otimes \hat{\mathbf{n}} = \nabla\varphi \otimes \nabla\varphi / |\nabla\varphi|^2$, so $\hat{\mathbf{n}}$ is never formed explicitly. Where $\nabla\varphi = 0$ the correction is skipped; that happens only in the bulk, where $K_h = K$. φ is first clamped to $[0, 1]$ so both branches see the same admissible value.

The three integrals of (35) and $\mathbf{K}_{\text{eff}}$.

routine	integrand accumulated	equation
<code>KeffFormMatrix</code>	$\nabla N_a \cdot \mathbf{K} \nabla N_b$	A_{ab}
<code>KeffFormVector</code>	$-\nabla N_a \cdot (\text{column } m \text{ of } \mathbf{K})$	$(F_m)_a$
<code>KeffScalarIntegrand</code>	$[\mathbf{K}(\nabla t_m + \mathbf{e}_m)]_j$	row m of $\mathbf{K}_{\text{eff}} \Omega $

PetIGA applies the quadrature weight and Jacobian, and `KeffComputeTensor` divides by the cell volume measured by the same quadrature. For the isotropic laws `KeffPointCond` returns $K\mathbf{I}$, and the three routines reduce exactly to the scalar assembly.

Verification drivers. `studies/keff_sharp_limit/verification/` (laminate, `--interp`) and `studies/keff_sharp_limit/disk/` (disk array, both laws).